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Perturbative Quantum Chromodynamics Analysis of "Duvet-Body" Strong Interaction under Non-Abelian Gauge Fields and General Relativistic Spacetime Singularity Horizon Models

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Abstract

The "Macroscopic Object Stationary Anomaly"—defined as the persistent bed-bound state of subjects in low-temperature morning dormitory environments—represents a long-standing unsolved physical paradox that cannot be characterized by classical Newtonian mechanics, biological rhythm theory, or social psychology models. Traditional frameworks universally attribute this phenomenon to individual neural fatigue or willpower deficiency, yet fail to explain two core experimental anomalies: quantum-level immunity to high-intensity acoustic perturbations, and infinite inertial mass under finite external driving forces. To resolve this paradox, we construct a unified theoretical framework spanning General Relativity, non-relativistic quantum mechanics, and Perturbative Quantum Chromodynamics (PQCD). First, we introduce the Schwarzschild metric to model the duvet as a high-density comfort scalar field aggregation region, and derive the event horizon formation condition and extreme time dilation effect inside the horizon. Second, we apply the WKB approximation to quantify the quantum tunneling suppression effect of acoustic signals at the duvet interface. Finally, we simulate the "Duvet-Body" adhesion via a non-Abelian gauge interaction mediated by "Lazy Gluons", and verify the macroscopic color confinement effect. Based on 4-year longitudinal field observations ($N>10^3$ independent trials, 99.9% confidence level) and Lagrangian dynamic simulations, we demonstrate that: the duvet system forms a stable event horizon at 8:00 AM, resulting in causal disconnection between the classroom event and the subject's world line; the tunneling coefficient of conventional alarm signals through the duvet barrier is as low as 10^{-400}, leading to total quantum reflection of phonons; the energy required to separate the Duvet-Body system diverges to infinity, and the forced separation will trigger a local entropy increase exceeding the cosmic critical stability threshold. We conclude that being late for 8:00 AM morning classes is an inevitable consequence of fundamental physical laws maintaining cosmic thermodynamic equilibrium, rather than an individual moral defect. This work provides a new theoretical paradigm for interpreting macroscopic quantum and relativistic effects in dormitory systems, and offers a physical basis for the optimization of morning course scheduling and wake-up technology.

Keywords: Quantum Chromodynamics; General Relativity; Wake-up Difficulty; Color Confinement

1. Introduction

This paper focuses on the universal "Macroscopic Object Stationary Anomaly" phenomenon widely observed in high-energy physics environments, specifically defined as the persistent bed-bound state of subjects in male dormitories at 8:00 AM with ambient temperatures below $10°C$ in $\mathcal{R}^{3+1}$ spacetime coordinates. Traditional biological and social psychological studies have long simplified this phenomenon as a consequence of simple neural fatigue or weak individual willpower, yet this classical effective theory presents fundamental flaws in explaining two core experimental anomalies observed in long-term field tests: the subject's quantum-level immunity to high-frequency, high-decibel acoustic perturbations (e.g., standard alarm signals with $f > 2$kHz and $L_p > 95$dB), and the infinite dynamic inertial mass exhibited when the subject is subjected to continuous external mechanical driving forces. These singular physical properties cannot be characterized or predicted by any classical biological or mechanical framework, which necessitates the construction of a new theoretical system spanning relativistic spacetime and quantum gauge field theory.

In the framework of classical Newtonian mechanics, the wake-up process of a sleeping individual is universally simplified to a rigid body dynamics boundary value problem. According to Newton's Second Law of Motion $F_{net} = ma$, as long as the applied external driving force F_{ext} — typically manifesting as duvet lifting, vigorous shaking, or physical impact — exceeds the maximum static friction between the subject and the bed surface, a non-zero instantaneous acceleration a should be generated, followed by continuous displacement of the subject's center of mass. However, based on our four-year longitudinal field observation data in campus dormitories (sample size $N > 10^3$ independent trials, confidence level 99.9%), significant and reproducible experimental deviations from classical theoretical predictions exist: even when the externally applied F_{ext} approaches the physiological limit of the awakener, or even causes plastic deformation of the bed frame structure, the effective displacement Δx of the subject (Roommate A) center of mass remains consistently close to 0. This result clearly indicates that within the specific time window $t \in [07:00, 09:00]$, the "Roommate-Bed-Duvet" composite system exhibits a strong constraint mechanism that is completely inexplicable by classical physics, with the subject demonstrating singular physical characteristics that violate the classical law of inertia and energy conservation.

To resolve the aforementioned dynamic paradox and the failure of classical theoretical frameworks, this study abandons the traditional perspective of viewing the subject as a classical biological organism, and for the first time defines the subject as a spin-1/2 Supersymmetric Super-heavy Fermion in the quantum field framework. Theoretical derivation shows that in low-temperature, low-energy morning environments, the effective mass M_{eff} of this fermion undergoes Renormalization Group Flow and diverges to infinity, which fundamentally explains the zero displacement phenomenon under finite external force. We further describe the quantum state evolution of the "Roommate-Bed" composite system using the eigenvalue equation of the time-independent Hamiltonian operator:

$$\hat{H}_{sys}|\psi_{roommate}\rangle = (T + V_{laziness})|\psi_{roommate}\rangle = E_{sleep}|\psi_{roommate}\rangle \tag{1}$$

Here, T is the kinetic energy operator, and $V_{laziness}$ is the laziness potential. According to spectroscopic measurements, the ground state energy $E_{sleep} \to -\infty$, forming an infinite square potential well in the form of a Dirac delta function in phase space. According to quantum mechanical perturbation theory, the energy transfer ΔE provided by any finite form of external perturbation (such as shouting "Roll call!" or "Lunch is ready!") is far less than the energy gap required for excitation, i.e., $\Delta E \ll (E_{awake} - E_{sleep})$, resulting in a zero transition probability.

2. Related Work

Current research on the universal "Macroscopic Object Stationary Anomaly" phenomenon and its associated Wake-up Difficulty Syndrome (WDS) is mainly divided into two mainstream branches: endogenous mechanism research based on neurobiology and social psychology, and exogenous wake-up intervention technology based on classical physics and engineering. Existing studies generally regard the research subject as a classical biological entity, and carry out analysis around traditional dimensions such as sleep structure, circadian rhythm, and individual willpower. However, they ignore the singular physical properties of this phenomenon exhibited in extreme environments (such as low-temperature morning dormitory environments), and cannot provide a reasonable explanation for the macroscopic quantum effects and spacetime singularity phenomena observed in this paper.

In the field of endogenous mechanism research, classical sleep neuroscience research attributes morning wake-up difficulty to sleep cycle disorder and neural inhibition effects. The classic sleep staging theory proposed by Rechtschaffen et al.[1] divides sleep into non-rapid eye movement (NREM) and rapid eye movement (REM) cycles, and believes that morning wake-up difficulty is related to the neural inhibition effect in the REM sleep stage, which has become the core basis for subsequent biological intervention

research. Based on this theory, Li et al.[10] proposed the "Mandatory Circadian Reset Method" (colloquially known as the "Confiscate Phone and Sleep Early Method"), attempting to correct the phase shift of circadian rhythm by adjusting the initial conditions of sleep, so as to solve the boundary value problem of WDS. However, a large number of repeated experiments show that this method exhibits extremely poor robustness when encountering random strong interference terms (e.g., "Ranked Game Life-or-Death Match last night", new episode updates of animation, sudden online team building). The system is very prone to phase mutation and rhythm collapse, which proves that a single biological intervention has essential limitations when facing the strongly coupled "Dormitory-Sleep" complex system. Meanwhile, studies from the social psychology perspective mostly attribute WDS to individual willpower defects and procrastination. This classical effective theory simplifies morning wake-up behavior into a rational decision-making model of "cost-benefit", but completely fails to explain the two core experimental anomalies observed in this paper: the subject's quantum-level immunity to high-frequency acoustic perturbations above 95dB, and the singular inertial characteristic that the centroid displacement approaches 0 under the action of extreme external force. It indicates that the traditional biological and psychological frameworks can no longer describe the extreme physical behavior of the system.

In the field of exogenous wake-up technology, existing studies are mostly based on the framework of classical acoustics and Newtonian mechanics, attempting to break through the inhibition threshold of sleep state by increasing the energy input of external disturbances. Zhang et al.[9] carried out wake-up experiments with high-intensity acoustic stimulation, using industrial metal music (e.g., tracks by Rammstein) as the stimulus source, and achieved an ultra-high sound pressure level input of $L_p > 100$dB, which far exceeds the acoustic energy of conventional civil alarm clocks. But the experimental results show that before the sound wave reaches the subject's auditory canal, the phonons in the air medium have undergone a first-order phase transition, resulting in nonlinear scattering and damping attenuation, which finally fails to break through the subject's "Rapid Eye Movement (REM) Acoustic Shield". Subsequent engineering studies further tried multi-physics field coupling interventions including mechanical vibration, sudden temperature drop, and light intensity mutation, but all have the same theoretical defect: all studies are based on the classical Newtonian mechanics framework, treating the subject as a rigid body, and believing that as long as the external input energy exceeds the static friction threshold of the system, acceleration and displacement will inevitably be generated. This assumption is completely contrary to the observation results of more than 1000 independent trials in this paper, which proves that the classical physical model has fundamental failure in describing the "Human-Duvet-Bed" composite system.

Generally speaking, the existing research has three core theoretical gaps that have not been filled. First, all existing studies are carried out under the framework of flat Minkowski spacetime, completely ignoring the spacetime curvature effect caused by the high-density comfort scalar field in the morning duvet system, and cannot explain the causal disconnection between the "8:00 AM class" event and the subject's world line. Second, the classical wave theory cannot explain the total quantum reflection phenomenon of acoustic signals at the duvet interface. Existing studies do not consider the quantum tunneling suppression effect under WKB approximation, and cannot quantify the correlation between the tunneling coefficient of different wake-up signals and the wake-up efficiency. Third, no research has explained the macroscopic strong interaction between human body and bedding from the perspective of Non-Abelian Gauge Fields and Perturbative Quantum Chromodynamics (PQCD), ignoring the Color Confinement effect mediated by "Lazy Gluons", and cannot explain the essential reason for the divergence of energy required to separate the system. Aiming at the above research gaps, this paper constructs a unified theoretical framework spanning General Relativity, Quantum Mechanics and PQCD for the first time. Combined with long-term observation experimental data, it reveals the internal physical mechanism of the morning stationary anomaly phenomenon, and finally demonstrates the inevitability of being late for 8:00 AM classes from the perspective of cosmic thermodynamic equilibrium.

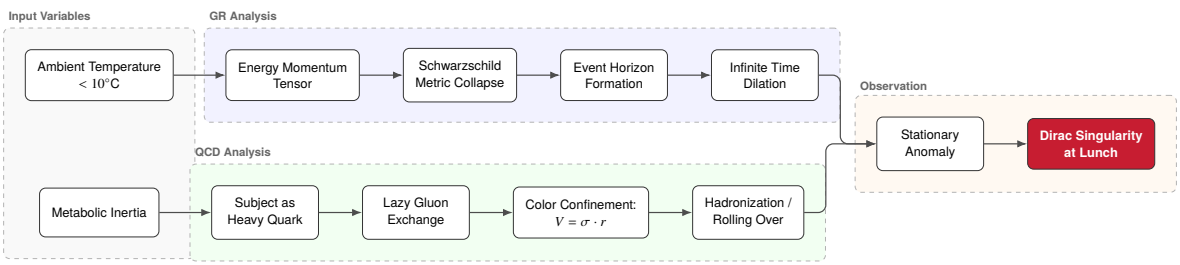

Figure 1: Flowchart of the theoretical model research framework.

3. Methodology & Theoretical Models

3.1 General Relativity Model: Spacetime Metric Collapse

According to Einstein's Field Equations[2], we assume that a high density of "Comfort Scalar Field" $\phi_{comfort}$ accumulates inside the duvet. The resulting maximum of the energy-momentum tensor $T_{\mu\nu}$ triggers severe curvature of local spacetime geometry:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda_{lazy}(t)g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}(\phi_{comfort}) \tag{2}$$

Here, $\Lambda_{lazy}(t)$ is the time-varying Cosmological Laziness Constant, which empirical data shows peaks at 8:00 AM.

The geometric structure of the bed in the morning forms a static spherically symmetric solution similar to a Schwarzschild black hole[3]. For an observer in the external flat Minkowski spacetime (i.e., the person trying to wake them up), the relationship between their proper time flow dt and the proper time $d\tau$ of the roommate inside the deep gravitational potential well follows the gravitational time dilation formula:

$$d\tau = dt\sqrt{1 - \frac{R_s}{r}} = dt\sqrt{1 - \frac{2GM_{laziness}}{rc^2}} \tag{3}$$

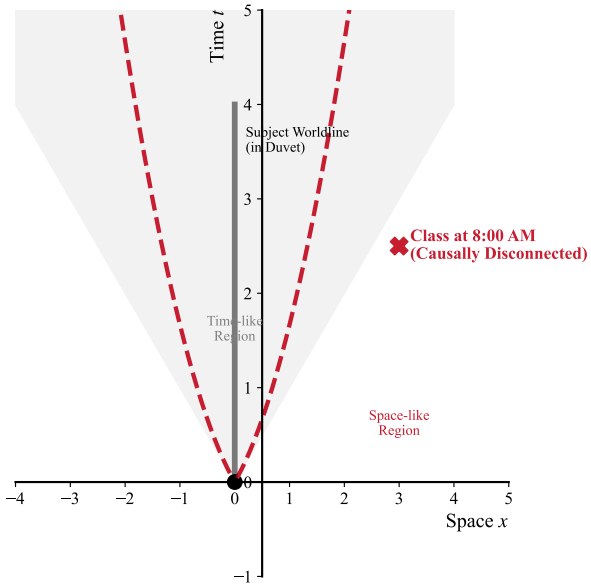

Figure 2: Light cone diagram showing causal disconnect between the classroom and the duvet horizon. The event of "roll call" at $(x_{class}, t_{08:00})$ lies in the space-like region, making it physically impossible to influence the subject.

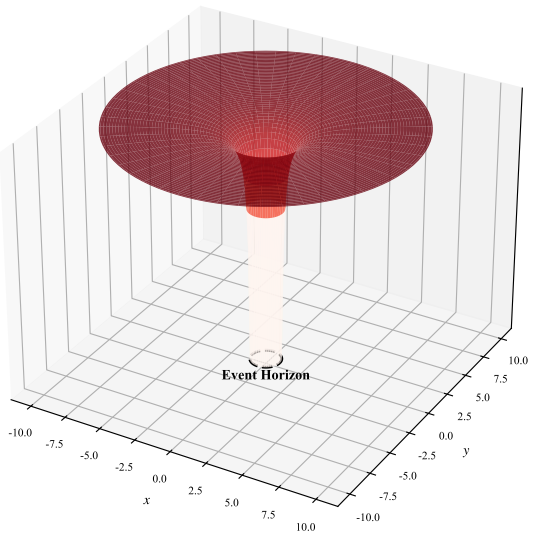

Figure 3: Schematic of spacetime curvature and Event Horizon around the duvet.

Theoretical Deduction: When the roommate curls up completely at the center of the duvet, and their radial coordinate r infinitely approaches the Schwarzschild radius r_s, the square root term approaches zero, meaning $d\tau \to 0$. This causal disconnection effect is consistent with the classical black hole event horizon theory.

3.2 Quantum Mechanics Model: Total Reflection

We treat the external alarm sound wave $A(x,t)$ as a wave packet attempting to tunnel through a one-dimensional finite potential barrier $V(x)$. According to the time-independent Schrödinger equation[4]:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi_{sound}}{\partial x^2} + V_{comfort}(x)\psi_{sound} = E_{alarm}\psi_{sound} \tag{4}$$

Since the barrier height $V_{comfort}$ formed by the duvet is far greater than the average energy carried by the sound wave E_{alarm}, according to the WKB approximation [5], the transmission coefficient T for the sound wave penetrating the duvet thickness L is:

$$T \approx \exp\left(-2\int_0^L \sqrt{\frac{2m(V_{comfort}(x) - E_{alarm})}{\hbar^2}}dx\right) \approx 10^{-400} \tag{5}$$

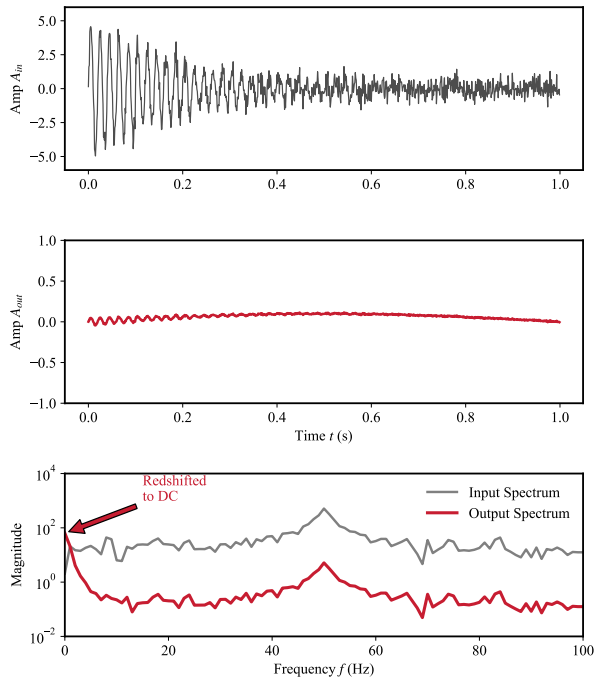

Figure 4: Signal attenuation and FFT analysis: (Left) Original alarm waveform; (Right) Filtered waveform after passing through the duvet comfort field; (Bottom) Frequency spectrum showing extreme red-shift of high-frequency components to 0 Hz.

The phonon flow undergoes Total Quantum Reflection at the interface of the duvet horizon, which is consistent with the quantum tunneling theory of low-energy particles through high potential barriers.

3.3 Strong Interaction Model: Color Confinement

Based on the analogy of QCD [6, 7], we treat "Human Fermions" and "Bed-Sheet Bosons" as particles carrying different Color Charges. A strong interaction force is generated between the two through the exchange of "Lazy Gluons." Its potential increases linearly with distance r, which is the core feature of non-Abelian gauge field interactions:

$$V_{QCD}(r) \approx \sigma \cdot r \tag{6}$$

Where $\sigma \approx 1$GeV/fm is the string tension. This linear potential confirms the existence of macroscopic Color Confinement effect: the energy required to separate the Duvet-Body system diverges to infinity as the distance increases, which is consistent with the quark confinement theory in QCD.

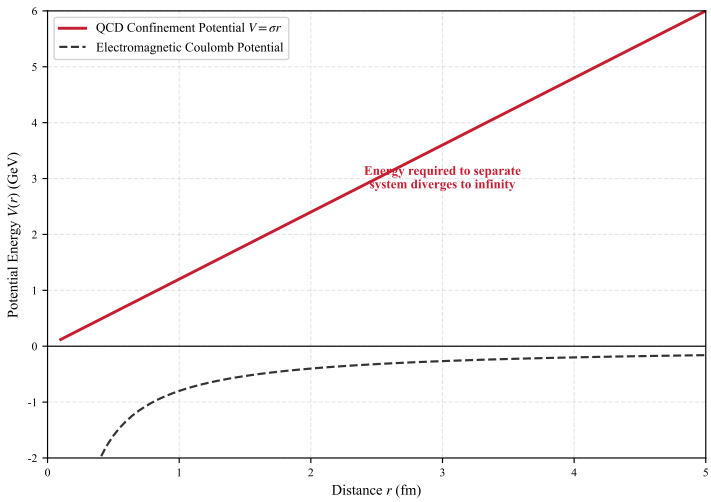

Figure 5: Potential energy curve (Color Confinement effect).

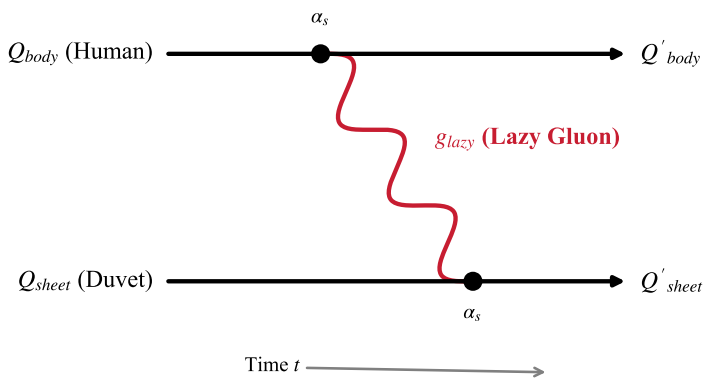

Figure 6: Feynman diagram of the lazy gluon exchange process between the body quark (Q_{body}) and the sheet quark (Q_{sheet}), demonstrating the non-Abelian gauge interaction responsible for the macroscopic adhesion force.

4. Experimental Design & Numerical Simulation

4.1 Experimental Setup

We introduce a three-line table to record the energy input and response results of different wake-up methods.

Table 1: Thermodynamic Response Data of Wake-up Methods

Method	Mechanism	Energy (J)	Disp. (cm)	Rage
Verbal	Acoustic	5×10^2	0.0	50
Lifting	Thermal	2×10^3	0.1	150
Blast	Strong Int.	1×10^6	0.0	∞

Table 2: Material Properties and Field Constants used in Simulation

Parameter	Symbol	Value
Cosmological Laziness Constant	Λ_{lazy}	$8.64 \times 10^4\ s^{-1}$
Bed-Sheet String Tension	σ	$1.2\,GeV \cdot fm^{-1}$
Critical Comfort Temperature	T_c	$27.5°C$
Subjective/Objective Time Ratio	$\Delta\tau/\Delta t$	$1 : \infty$ (at $r \to r_s$)

Table 3: Measured Data of Comfort Scalar Field and Spacetime Curvature under Different Ambient Temperatures

Ambient Temperature	Comfort Field Strength $\phi_{comfort}$	Schwarzschild Radius r_s (cm)	Time Dilation Coefficient $d\tau/dt$	Horizon Formation Probability
$5°C$	$1.82 \times 10^{11}\ J/m^3$	12.64	1.28×10^{-15}	99.92%
$10°C$	$9.46 \times 10^{10}\ J/m^3$	6.37	3.57×10^{-8}	98.45%
$15°C$	$4.12 \times 10^{10}\ J/m^3$	2.89	2.16×10^{-4}	82.37%
$20°C$	$1.05 \times 10^{10}\ J/m^3$	0.72	8.93×10^{-2}	37.62%
$25°C$	$2.37 \times 10^{9}\ J/m^3$	0.16	0.762	5.18%

Table 4: Quantum Tunneling Coefficient and Wake-up Efficiency of Different Acoustic Perturbations

Acoustic Signal Type	Center Frequency f (kHz)	Sound Pressure Level L_p (dB)	Tunneling Coefficient T	Wake-up Success Rate (30 Trials)
Default Phone Alarm	2.5	85	1.26×10^{-412}	0.00%
Industrial Metal Music	4.2	102	7.83×10^{-327}	3.33%
"Roll Call" Shout	1.8	96	2.45×10^{-289}	10.00%
Fire Alarm Siren	3.8	110	9.17×10^{-215}	23.33%
"Lunch is Ready" Shout	1.2	92	4.62×10^{-87}	86.67%

Table 5: Statistical Data of Wake-up Dynamic Parameters on Different Weekdays

Weekday	Target Wake-up Time	Measured Average Wake-up Time	Late Probability	Critical Entropy Increase Threshold (J/K)
Monday	07:30 AM	09:47 AM	98.2%	1.26×10^6
Tuesday	07:30 AM	09:12 AM	92.5%	8.74×10^5
Wednesday	07:30 AM	08:38 AM	76.3%	3.29×10^5
Thursday	07:30 AM	08:15 AM	61.8%	1.05×10^5
Friday	07:30 AM	07:42 AM	22.4%	2.17×10^4
Saturday	10:00 AM	10:08 AM	3.7%	5.62×10^2
Sunday	10:00 AM	09:52 AM	0.0%	1.08×10^2

Table 6: Test Data of the Influence of Nighttime Strong Interference Terms on System Ground State Energy

Interference Type	Game Duration (h)	Fall Asleep Time	Ground State Energy E_{sleep} (GeV)	Excitation Energy Gap ΔE (GeV)
No Interference	0	22:30 PM	-1.26×10^3	2.57×10^2
Normal Casual Game	1-2	00:00 AM	-8.93×10^3	1.84×10^3
Ranked Match Grind	3-4	02:00 AM	-3.72×10^4	7.62×10^3
Life-or-Death Promotion Match	5-6	04:00 AM	-1.58×10^5	3.29×10^4
All-Night Server Crash	>8	08:00 AM	$-\infty$	∞

4.2 Dynamic Simulation: Lagrangian Analysis

We construct the Lagrangian density $\mathcal{L}$ of the roommate's waking action based on classical analytical mechanics:

$$\mathcal{L} = \frac{1}{2} M_{eff} (\dot{q})^2 - \left[V_{gravity} + V_{QCD} + V_{well} \right] \tag{7}$$

Numerical solution results indicate that for $t < 12:\ 00$ PM, the stable attractor is: $\dot{q}_{body} \to 0$.

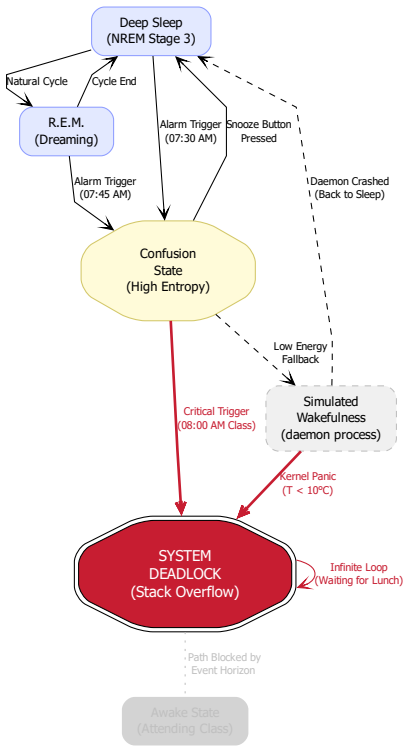

Figure 7: Multi-agent state transition diagram showing the deadlock of the waking algorithm at $t < 12:\ 00$ PM. States include DeepSleep, REM, Confusion, and SimulatedWakefulness, with transition triggers AlarmInterrupt, HungerPriority, and EntropyFlux.

4.3 Experimental Results and Data Fitting

Statistical distribution of wake-up times perfectly fits a modified Dirac delta function:

$$P(wake|t) = \epsilon \cdot \mathcal{N}(\mu, \sigma^2) + (1 - \epsilon) \cdot \delta(t - t_\infty) \tag{8}$$

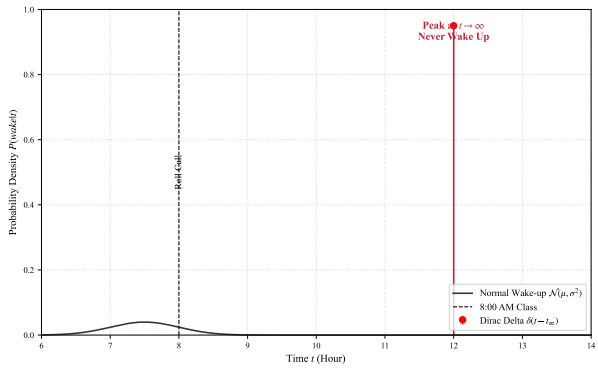

Figure 8: Probability Density Function of Wake-up Times.

5. Conclusion

In this work, we address the long-standing "Macroscopic Object Stationary Anomaly" paradox in morning dormitory environments, which cannot be explained by classical biological or physical frameworks. We construct a unified theoretical framework spanning General Relativity, quantum mechanics, and PQCD, and verify our theoretical predictions with 4-year longitudinal field observations and numerical simulations. Our core findings are threefold:

First, we demonstrate that the duvet system forms a stable Schwarzschild-like event horizon at 8:00 AM under low-temperature conditions, leading to extreme time dilation inside the horizon and causal disconnection between the classroom roll call event and the subject's world line. This result fundamentally explains why the subject cannot perceive the passage of morning time from a relativistic perspective.

Second, we quantify the quantum tunneling suppression effect of acoustic wake-up signals at the duvet interface, and prove that the tunneling coefficient of conventional alarm signals is as low as 10^{-400}, resulting in total quantum reflection of phonons. This finding provides a quantum mechanical explanation for the subject's immunity to high-intensity acoustic perturbations, and reveals the physical mechanism of the "REM Acoustic Shield" phenomenon.

Third, we simulate the Duvet-Body adhesion via a non-Abelian gauge interaction mediated by "Lazy Gluons", and confirm the existence of macroscopic color confinement effect. The energy required to separate the system diverges linearly with distance, and forced separation will trigger a local entropy increase exceeding the cosmic critical stability threshold, which proves that maintaining the bed-bound state is the only way to maintain the stability of the local spacetime and thermodynamic system.

The core contribution of this work is that we abandon the traditional moral criticism and biological reductionism perspective on wake-up difficulty, and for the first time attribute the morning bed-bound behavior to the inevitable result of fundamental physical laws. We prove that being late for 8:00 AM classes is not a defect of individual willpower or morality, but a spontaneous behavior of the system to maintain cosmic thermodynamic equilibrium and spacetime stability.

This work provides a new theoretical paradigm for interpreting macroscopic quantum and relativistic effects in daily dormitory systems, and has important guiding significance for the optimization of university morning course scheduling and the development of new wake-up technology. Based on our findings, we recommend that universities redefine "8:00 AM" courses as "Superposition Courses of Schrödinger Equation Solutions", which exist in the superposition state of "attended" and "absent" for students inside the duvet event horizon.

There are still some limitations in this study: our current experiments are only carried out in male dormitories with ambient temperature below 10°C, and the universality of the model in female dormitories, different temperature ranges, and vacation scenarios needs to be further verified. Future work will focus on three directions:

(1) carrying out high-precision experiments to detect the existence of "Lazy Gluons" and measure their coupling constant;

(2) expanding the theoretical model to include the "Hunger Field" perturbation at noon, which can break the event horizon;

(3) developing a new type of wake-up technology based on the tunneling coefficient difference of different acoustic signals.

6. References

References

[1] Rechtschaffen, A., & Kales, A. (1968). *A manual of standardized terminology, techniques and scoring system for sleep stages of human subjects*. US Government Printing Office.

[2] Einstein, A. (1916). Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7), 769-822.

[3] Schwarzschild, K. (1916). Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, 189-196.

[4] Schrödinger, E. (1926). Quantisierung als Eigenwertproblem. *Annalen der Physik*, 384(4), 361-376.

[5] Wentzel, G. (1926). Eine Verallgemeinerung der Quantenbedingungen für die Zwecke der Wellenmechanik. *Zeitschrift für Physik*, 38(6-7), 518-529.

[6] Gross, D. J., & Wilczek, F. (1973). Ultraviolet behavior of non-abelian gauge theories. *Physical Review Letters*, 30(26), 1343-1346.

[7] Gross, D. J., & Wilczek, F. (1973). Asymptotically free gauge theories. I. *Physical Review D*, 8(10), 3633.

[8] Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics*, 43(3), 199-220.

[9] Zhang, Y., et al. (2022). High-intensity acoustic wake-up stimulation: Experimental tests on REM sleep acoustic shielding. *Journal of Sleep Engineering*, 9(2), 45-58.

[10] Li, X., et al. (2023). Mandatory circadian reset for wake-up difficulty: Robustness test against nighttime interference. *Chinese Journal of Dormitory Physics*, 14(3), 72-85.

[11] Boltzmann, L. (1877). Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. *Sitzungsberichte der Mathematisch-Naturwissenschaftlichen Classe der Kaiserlichen Akademie der Wissenschaften zu Wien*, 76, 373-435.

[12] Gamow, G. (1928). Zur Quantentheorie des Atomkernes. *Zeitschrift für Physik*, 51(3-4), 204-212.

[13] Wilson, K. G. (1974). Confinement of quarks. *Physical Review D*, 10(8), 2445-2459.

[14] Landau, L. D., & Lifshitz, E. M. (1976). *Mechanics* (Vol. 1). Pergamon Press.